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Limits to Arbitrage and Hedging: Evidence from Commodity Markets

Contents

1 Big picture

- **Question.** Can limits to arbitrage in commodity futures markets make hedging costly for producers, and can this financial friction affect both futures risk premia and spot commodity prices?
- **Setting.** The paper studies oil and gas related commodity markets: crude oil, heating oil, gasoline, and natural gas. The empirical sample combines NYMEX futures and spot-price proxies, producer balance-sheet and market default-risk measures, EDGAR hedging disclosures, CFTC position data, inventories, and broker-dealer balance-sheet growth.
- **Core mechanism.** Producers naturally want to short futures to hedge future production. Speculators must take the opposite long side. If speculators are capital constrained, their demand curve for futures is downward sloping. More producer hedging demand then depresses futures prices and raises futures expected returns.
- **Real-side link.** Hedging is not only a financial trade. If hedging becomes expensive, producers hedge less and carry less inventory. Lower inventory means more current supply in the spot market, which depresses current spot prices relative to future spot prices.
- **Main theoretical prediction.** Producer hedging demand and speculator capital constraints increase both the commodity futures risk premium and the expected percentage change in the spot price.
- **Main empirical proxy.** The paper uses producer default risk as a proxy for fundamental hedging demand. The logic is that managers and firms have stronger incentives to hedge when financial distress is more likely.
- **Micro validation.** Firm-level EDGAR disclosures show that oil and gas producers commonly hedge and that firms with higher default risk are more likely to report significant short crude-oil derivative positions.
- **Aggregate result.** Higher aggregate producer default risk predicts higher next-quarter futures returns and higher next-quarter spot price changes across the four oil and gas commodities.
- **Limits-to-arbitrage result.** The predictive relation is stronger when commodity return volatility is high and when broker-dealer balance sheets shrink, consistent with a tighter speculator capital constraint.
- **Economic takeaway.** Limits to arbitrage in financial markets create limits to hedging for real-sector firms. In commodity markets, this can move not only risk premia in futures markets, but also real spot prices through producers' inventory choices.

2 Data and measurement: key objects

2.1 Commodity markets and sample

The empirical analysis focuses on four energy commodities: crude oil, heating oil, gasoline, and natural gas. This choice is driven by data availability. Publicly traded producer firms are numerous enough in oil and gas to construct aggregate producer default-risk measures.

The main producer sample consists of firms with SIC code 1311, Crude Petroleum and Natural Gas Extraction. The broad producer accounting sample runs from 1979Q3 to 2010Q4. The micro hedging-disclosure sample uses EDGAR annual and quarterly filings from 2000Q1 to 2010Q4, after the introduction of FAS 133 disclosure requirements.

2.2 Futures returns and spot-price changes

The paper constructs rolling commodity futures excess returns using the nearest-to-maturity contract that does not expire during the next month:

$$r^f_{t,t+1} = \frac{F_{t+1,T} - F_{t,T}}{F_{t,T}}. \tag{1}$$

Interpretation. The return is the gain from buying a futures contract at the end of month t and marking the same contract to market at the end of month $t+1$. Quarterly returns compound three monthly returns.

The spot-price change is proxied using the nearest-to-expiration futures contract:

$$\Delta s_{t,t+1} = \frac{F_{t+1,t+2} - F_{t,t+1}}{F_{t,t+1}}. \tag{2}$$

Interpretation. Since true spot-price data may be less comparable across commodities, the paper uses the nearest futures contract as a standardized spot-price proxy. The key prediction is that the same force that raises the futures risk premium should also raise expected spot-price changes.

2.3 Producer default risk as hedging demand

The key empirical proxy for fundamental producer hedging demand is producer default risk. The paper uses two measures.

Zmijewski-score

The balance-sheet-based proxy is the Zmijewski-score:

$$\text{Zmijewski-score} = -4.3 - 4.5\frac{\text{Net Income}}{\text{Total Assets}} + 5.7\frac{\text{Total Debt}}{\text{Total Assets}} - 0.004\frac{\text{Current Assets}}{\text{Current Liabilities}}. \tag{3}$$

Interpretation. A higher score corresponds to greater financial distress risk. Lower profitability and higher leverage increase the score. This proxy is useful because it is available over a long sample.

KMV expected default frequency

The market-based proxy is KMV's expected default frequency, EDF:

$$EDF = \Phi\left(-\frac{\ln(V/F) + (\mu - 0.5\sigma_V^2)T}{\sigma_V\sqrt{T}}\right), \tag{4}$$

where V is the total market value of the firm, F is the face value of debt, σ_V is asset volatility, μ is expected asset return, and T is the horizon.

Interpretation. EDF uses market information and balance-sheet information to estimate the probability that the firm defaults within the horizon. It is conceptually close to the distress-risk channel emphasized by the paper.

2.4 Micro hedging disclosures

The authors hand-collect producer hedging information from EDGAR filings. For each firm-quarter, they code whether the firm has significant short crude-oil derivatives positions. The important distinction is between a firm that merely says it may use derivatives and a firm that reports economically significant short hedges.

Why this matters. The paper's aggregate tests require default risk to be a proxy for hedging demand. The micro disclosures provide a direct validation: firms with higher default risk are more likely to hedge.

2.5 Speculator capital and limits to arbitrage

The paper measures speculator risk tolerance using broker-dealer balance-sheet growth relative to household asset growth. The idea comes from the limits-to-arbitrage literature: when broker-dealer balance sheets expand, intermediary capital is more abundant; when they shrink, speculative capital is tighter.

Let BD_t denote broker-dealer asset growth relative to household asset growth. A high BD_t means high speculator risk tolerance, or weaker limits to arbitrage. A low BD_t means tight intermediary capital, or stronger limits to arbitrage.

3 Model and empirical specifications

3.1 Spot demand and commodity supply

The model starts with an inverse demand curve for the commodity:

$$S_t = \omega \left(\frac{C_t}{Q_t} \right)^{1/\varepsilon}, \tag{5}$$

where S_t is the spot price, Q_t is equilibrium commodity supply, C_t is demand for other goods, and ε controls demand elasticity.

Interpretation. Spot prices are high when commodity supply Q_t is low relative to demand C_t. This is the channel through which inventory decisions affect spot prices.

3.2 Producer objective

A producer chooses inventory i and a short futures position h_p. The manager maximizes firm value minus a penalty for earnings variance:

$$\max_{i,h_p} \quad S_0(g_0 - i) + \mathbb{E}\left[\Lambda\{S_1((1-\delta)i + g_1) + h_p(F - S_1)\}\right] \tag{6}$$

$$- \frac{\gamma_p}{2}\operatorname{Var}\left[S_1((1-\delta)i + g_1) + h_p(F - S_1)\right] \tag{7}$$

$$\text{subject to} \quad i \geq 0. \tag{8}$$

Interpretation. The producer sells current output, stores some inventory, and hedges future exposure by shorting futures. The parameter γ_p captures fundamental hedging demand. A higher γ_p means the manager dislikes unhedged commodity-price risk more intensely.

The producer's desired short futures position satisfies:

$$h_p^* = i^*(1-\delta) + g_1 - \frac{\mathbb{E}[\Lambda(S_1 - F)]}{\gamma_p \sigma_S^2}. \tag{9}$$

Interpretation. The first term is the physical exposure from future inventory and production. The second term reflects the cost of hedging. If futures prices are low relative to expected future spot prices, hedging is expensive, so producers hedge less than their full physical exposure.

3.3 Speculators and market clearing

Speculators take the long side of the futures positions that producers want to short. They are modeled as specialized arbitrageurs with limited capital or risk-bearing capacity. A higher γ_s means stronger speculator risk aversion or tighter capital constraints.

The futures market clears when producer short positions equal speculator long positions. This market-clearing condition is the source of price pressure: when producers want to hedge more, constrained speculators demand compensation to absorb that risk.

3.4 Futures risk premium

The key pricing equation is:

$$\mathbb{E}\left[\frac{S_1 - F}{F}\right] = -(1+r)\operatorname{Corr}(\Lambda, S_1)\operatorname{Std}(\Lambda)\sigma_f + \frac{\gamma_p\gamma_s}{\gamma_p + \gamma_s}(1+r)\sigma_f^2 F Q_1. \tag{10}$$

Interpretation. The first term is the standard asset-pricing component: futures payoffs that covary with the representative investor's marginal utility should earn a risk adjustment. The second term is the hedging-pressure component. It rises when producer hedging demand γ_p is high, when speculator constraints γ_s are high, when futures risk σ_f^2 is high, and when the dollar amount of exposure FQ_1 is large.

This equation is the central bridge between theory and empirics. The paper's empirical work asks whether proxies for γ_p and γ_s forecast futures returns and spot-price changes in the way the model predicts.

3.5 Proposition 1: main comparative statics

The model implies:

$$\frac{d}{d\gamma_p}\mathbb{E}\left[\frac{S_1 - F}{F}\right] > 0, \tag{11}$$

$$\frac{d}{d\gamma_s}\mathbb{E}\left[\frac{S_1 - F}{F}\right] > 0. \tag{12}$$

When there is no stock-out, the expected percentage spot-price change also rises with producer hedging demand and speculator constraints:

$$\frac{d}{d\gamma_p}\mathbb{E}\left[\frac{S_1 - S_0}{S_0}\right] > 0, \tag{13}$$

$$\frac{d}{d\gamma_s}\mathbb{E}\left[\frac{S_1 - S_0}{S_0}\right] > 0. \tag{14}$$

Interpretation. More hedging demand or less speculator capital raises the cost of hedging. Producers respond by carrying less inventory. Less inventory means more current spot supply, lower current spot prices, and therefore higher expected future spot-price changes.

3.6 Baseline forecasting regression

The main aggregate test is:

$$\text{ExcessReturns}_{i,t+1} = \beta_i \text{DefRisk}_{i,t} + \text{ControlVariables}_t + u_{i,t+1}, \tag{15}$$

where i indexes commodities.

Interpretation. If producer default risk captures hedging demand, then β_i should be positive. A positive coefficient means that when producers are more distressed and more inclined to hedge, the next-quarter futures return is higher.

The spot-price test uses the same structure:

$$\Delta s_{i,t+1} = \beta_i \text{DefRisk}_{i,t} + \text{ControlVariables}_t + u_{i,t+1}. \tag{16}$$

Interpretation. The model predicts a similar positive sign for spot-price changes because producer inventory decisions link the futures-market friction to the spot market.

3.7 Volatility interaction

From the risk-premium equation, the hedging-pressure term is larger when futures return variance is high. The paper tests:

$$\text{ExcessReturns}_{i,t+1} = \beta_1 RV_{i,t} + \beta_2 \text{DefRisk}_{i,t} + \beta_3 RV_{i,t} \times \text{DefRisk}_{i,t} + \text{Controls}_t + u_{i,t+1}. \tag{17}$$

Interpretation. $RV_{i,t}$ is realized variance. The prediction is $\beta_3 > 0$: a given increase in producer hedging demand should have a bigger effect when the risk that speculators must absorb is larger.

3.8 Speculator-capital interaction

The paper also tests whether the default-risk effect is weaker when speculative capital is abundant:

$$\text{ExcessReturns}_{i,t+1} = \beta_1 BD_t + \beta_2 \text{DefRisk}_{i,t} + \beta_3 BD_t \times \text{DefRisk}_{i,t} + \text{Controls}_t + u_{i,t+1}. \tag{18}$$

Interpretation. Here BD_t is broker-dealer balance-sheet growth relative to household asset growth. Because high BD_t means loose speculator capital, the model predicts $\beta_3 < 0$. Hedging demand should matter less when arbitrage capital is plentiful.

3.9 Positions, inventory, and basis tests

The paper also studies whether default risk predicts objects closer to the mechanism:

$$\text{HedgerNetPos}_{i,t} = \beta \text{DefRisk}_{i,t} + \gamma_i \text{HedgerNetPos}_{i,t-1} + \text{Controls}_t + u_{i,t}, \tag{19}$$

$$\Delta\text{Inventory}_{i,t+1} = \beta \text{DefRisk}^{IV}_{i,t+1} + \sum_{j=1}^{3} \gamma_{i,j} \Delta\text{Inventory}_{i,t+1-j} + \text{Controls}_t + u_{i,t+1}, \tag{20}$$

$$\text{basis}_{i,t+1} = \beta \text{DefRisk}^{IV}_{i,t+1} + \gamma_i \text{basis}_{i,t} + \text{Controls}_t + u_{i,t+1}. \tag{21}$$

Interpretation. Higher default risk should be associated with more net short hedger positions, lower inventories, and a higher basis. The inventory and basis tests use an instrumental-variables approach based on persistent default risk to reduce the impact of contemporaneous commodity-demand shocks.

4 Identification and credibility

4.1 Main identification problem

The key challenge is that producer default risk is not randomly assigned. Default risk could be correlated with business-cycle conditions, expected commodity demand, supply shocks, or standard risk-premium variation. The paper therefore cannot rely on a simple predictive regression alone.

The identification strategy is cumulative: show that the proxy is connected to hedging at the micro level, then show that the aggregate pricing relation survives many controls and is stronger exactly when the model says it should be stronger.

4.2 Why default risk proxies hedging demand

The conceptual reason is that hedging becomes more valuable when distress costs are high. Managers may dislike default because it increases job-loss risk, relocation costs, lost human capital, and loss of firm-specific compensation. Firms may also hedge to reduce costly financial distress and external-finance costs.

The empirical support comes from EDGAR disclosures: producers with higher Zmijewski-scores and higher EDFs are more likely to report significant short crude-oil hedging. This is the paper's main validation of the default-risk proxy.

4.3 Business-cycle and risk-premium controls

The forecasting regressions include controls designed to absorb conventional sources of risk-premium variation:

- the aggregate default spread,
- the three-month Treasury bill rate,
- forecasts of GDP growth,
- the commodity futures basis,
- realized variance of futures returns,
- quarterly seasonal dummies.

Economic role. These controls help separate producer hedging demand from aggregate discount-rate variation and commodity-specific supply-demand conditions.

4.4 Hedger versus non-hedger partition

A strong robustness test splits producers into those that disclose significant hedging and those that do not. If producer default risk matters because it proxies hedging demand, then the default risk of hedgers should forecast futures returns, while the default risk of non-hedgers should not.

That is what the paper finds. The predictive power comes from hedging firms, not from non-hedging firms. This makes it harder to interpret the result as merely a generic default-risk or business-cycle effect.

4.5 Speculator capital interaction

The model says that producer hedging demand matters more when speculator capital is scarce. The paper tests this using broker-dealer balance-sheet growth. The interaction between default risk and broker-dealer growth is negative: when broker-dealer balance sheets grow, producer hedging demand has a smaller effect on futures returns and spot-price changes.

This interaction is important because it directly tests the "limits to arbitrage" part of the argument, not just the "hedging demand" part.

4.6 Inventory and spot-price evidence

The paper does not stop at futures returns. It also shows that producer default risk predicts spot-price changes and is associated with lower inventories. These results support the real-side channel: financial constraints in hedging markets affect producers' physical inventory choices, which affect spot-market prices.

4.7 Remaining limitations

The evidence is persuasive, but it is not a randomized experiment. Producer default risk may still contain information about expected future supply, demand, or sector conditions. The paper addresses this concern through controls, sample partitions, interactions, and inventory tests, but the interpretation remains model-guided rather than purely experimental.

5 Tables and figures

5.1 Figure 1: aggregate producer default risk

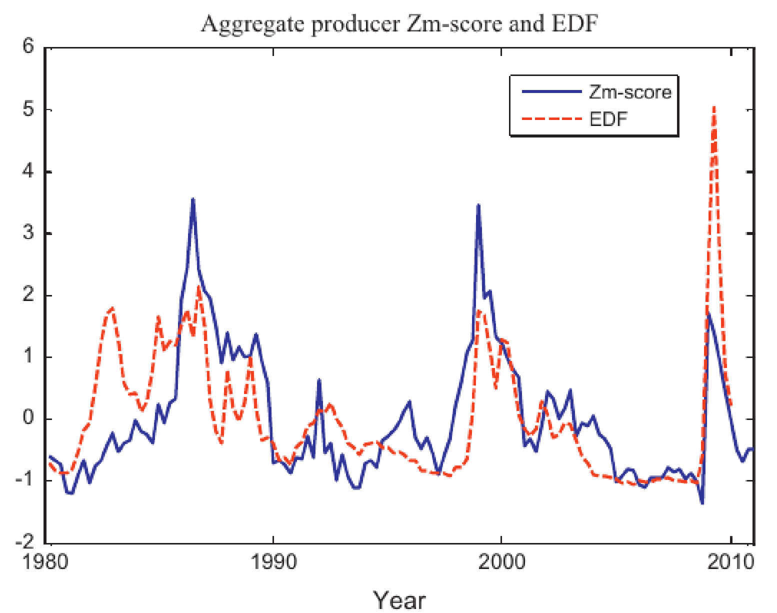

Fig. 1. Aggregate default risk measures. The graph shows the aggregate commodity default risk measures (the Zmijewski-score (Zm-score) and EDF) for Crude Oil and Natural Gas producers (SIC code 1311), measured quarterly. The Zm-score data are from 1979Q3 to 2010Q4. The EDF data are from 1980Q1 to 2009Q4. Both series are normalized to have zero mean and unit variance for ease of comparison.

Figure 1: Aggregate default-risk measures used as proxies for producer hedging demand.

Read: The figure plots aggregate Zmijewski-score and EDF measures for crude oil and natural gas producers. The two measures are normalized for comparison and are persistent over time.

Economic message. The paper needs a time-series proxy for hedging demand. Figure 1 shows that producer default risk varies substantially over the sample, creating time-series variation with which to test the theory.

5.2 Table 1 and Figure 2: micro evidence on hedging

Table 1
Micro-level hedging.

This table shows results using firm level hedging data collected from 10-Q reports for all firms with SIC code 1311 (Crude Oil and Natural Gas Extraction) in the period 2000Q1–2010Q4. For each firm-quarter, the fields given in Panel A were answered either 0 (No), 1 (Yes), or missing if no information was given. Panel A gives the summary statistics for these data. Panel B shows regressions where the dependent variable is the firm-quarter data point corresponding to the question "Does the firm currently have significant short crude oil derivatives positions?" Panel B shows the results from both OLS and dprobit regressions when measures of firm-quarter default risk are used as independent variables (the Zmijewski- and EDF-scores, respectively). Standard errors are robust and clustered by time. ** Denotes significance at the 5% level, and *** denotes significance at the 1% level.

Panel A: Summary statistics on micro-hedging

For each quarter and firm:	# Firm-quarter obs.	Fraction "yes"
Use derivatives?	2,400	88.0%
Futures or forwards?	547	47.7%
Swaps?	1,781	80.6%
Options?	1,800	81.8%
Significant short crude?	1,738	69.8%
Significant long crude?	964	1.0%

Panel B: Crude oil hedging vs. firm-level default risk measures

Dependent variable: "significant short crude?" (0 or 1 for each firm-quarter)

	Indep.var: Firm-quarter Zm-score					Indep.var: Firm-quarter log EDF-score				
	β	(S.E.)	Fixed effect	R^2_{adj}	N	β	(S.E.)	Fixed effect	R^2_{adj}	N
dprobit	0.053***	(0.014)	Time *f.e.*	6.1%	926	0.057***	(0.009)	Time *f.e.*	7.3%	1,100
	0.055***	(0.019)	Firm *f.e.*	18.6%	548	0.022***	(0.008)	Firm *f.e.*	20.6%	660
	0.058***	(0.020)	Time & firm *f.e.*	25.2%	548	0.058***	(0.016)	Time & firm *f.e.*	26.8%	660
OLS	0.055***	(0.014)	Time *f.e.*	5.0%	926	0.057***	(0.009)	Time *f.e.*	6.0%	1,100
	0.034***	(0.011)	Firm *f.e.*	29.8%	926	0.011**	(0.004)	Firm *f.e.*	31.6%	1,100
	0.041***	(0.011)	Time & firm *f.e.*	32.7%	926	0.036***	(0.009)	Time & firm *f.e.*	34.7%	1,100

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Figure 2: Micro-level hedging disclosures and the relation between firm default risk and hedging.

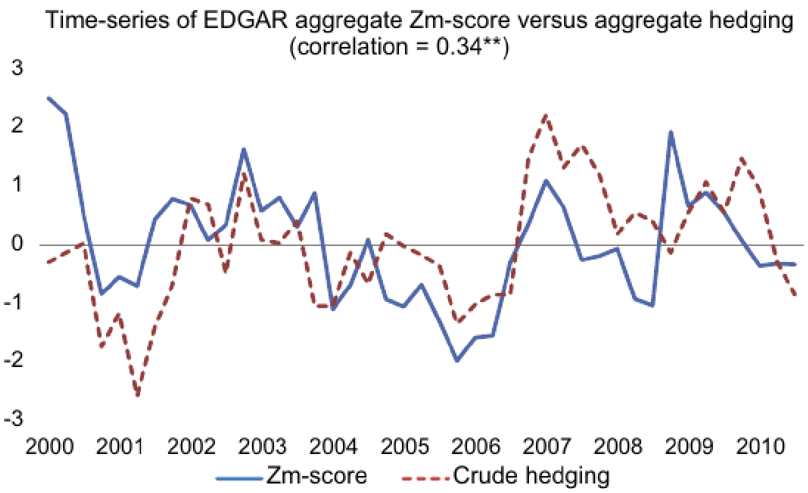

Fig. 2. Firm crude oil hedging and default risk. The top plot shows the average "Significant short crude derivatives positions" (see main text for definition) across firms in a given quarter for which 10-Q data on crude oil hedging are available versus the median Zmijewski-score across the same firms in the same quarter. The data period is 2000Q1–2010Q4. In order to facilitate easy comparison between these aggregate data series, both series have been normalized to have zero mean and unit variance.

Figure 3: Aggregate EDGAR hedging intensity and aggregate producer default risk.

Read: Table 1 shows that 88.0% of firm-quarters mention use of derivatives, 69.8% report significant short crude-oil positions, and only 1.0% report significant long crude-oil positions. Panel B shows that both Zmijewski-score and log EDF positively predict significant short hedging. Figure 2 shows a positive time-series relation between EDGAR-based aggregate hedging and the aggregate Zmijewski-score.

Economic message. This is the core measurement validation. Producers are mainly short hedgers, and distressed producers hedge more. Therefore producer default risk is a plausible proxy for fundamental hedging demand.

5.3 Table 2: summary statistics

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Table 2
Summary statistics.

Panel A shows summary statistics for the futures returns and the measures of commodity producer default risk, where producers are taken as firms with SIC code 1311 (Crude Oil and Natural Gas extraction). The longest sample (Heating Oil) is from 1979Q4 to 2010Q4. Crude Oil futures return data are from 1983Q3 to 2010Q4, Gasoline futures data are from 1985Q1 to 2010Q4, while Natural Gas futures data are from 1990Q3 to 2010Q4. The Expected default frequency (EDF) data are from 1979Q3 to 2009Q4, while the Zmijewski-score (Zm-score) data are from 1979Q3 to 2010Q3. Panel B shows summary statistics for the underlying set of firms in SIC code 1311 over the same sample period. Firms are required to have total assets larger than $10MM to be included in the sample. The percentiles are at the firm-year level. The aggregate leverage observations are annual. For the Zmijewski-score sample, there are 2,971 firm-year observations, whereas for the EDF sample from KMV, which contains larger firms on average, there are 892 firm-year observations. "VW" denotes "value-weighted."

Panel A: Aggregate

	Mean	St.dev.	AR(1)	*N*
Quarterly futures return:				
Crude Oil	3.75%	20.5%	−0.07	110
Heating Oil	4.16%	20.0%	−0.03	125
Gasoline	6.29%	21.4%	−0.11	104
Natural Gas	−0.28%	28.7%	0.08	82
Aggregate producer default risk measures:				
EDF	1.89%	1.70%	0.82	121
Zm	−2.60	0.334	0.76	125

Panel B: Firm-level

	5%-tile	25%-tile	50%-tile	75%-tile	95%-tile	Mean
Zmijewski-score sample:						
Firm size ($M)	9	44	194	946	7247	1701
Leverage	0%	0%	14%	31%	67%	20%
VW aggregate leverage	15%	19%	22%	29%	38%	25%
Zmijewski-score	−4.5	−3.8	−2.7	−1.7	1.2	−1.2
EDF-score sample:						
Firm size	15	122	542	2079	13877	2841
Leverage	0%	14%	27%	42%	77%	30%
VW aggregate leverage	18%	23%	30%	38%	49%	31%
EDF	0.1%	0.2%	0.6%	2.2%	16%	2.6%

Figure 4: Summary statistics for futures returns and producer default-risk measures.

Read: Oil futures returns are positive on average, with quarterly standard deviations around 20%. Natural gas has a slightly negative average return and higher volatility. Default-risk measures are persistent, with autoregressive coefficients around 0.76 to 0.82.

Economic message. The empirical setting has both meaningful commodity return variation and persistent producer distress variation. Persistence matters because default risk can forecast returns only if it has enough slow-moving variation to be observed before returns are realized.

5.4 Table 3: baseline futures and spot forecasting regressions

Table 3
Futures and spot return forecasting regressions.

Panel A shows the results from regressions of Crude Oil, Heating Oil, Gasoline, and Natural Gas futures returns on lagged default risk of oil and gas producers, as measured by the Expected default frequency (EDF) and Zmijewski-score (Zm) of firms with SIC code 1311. The controls in the regressions are lagged futures basis, aggregate default risk, GDP growth forecast (from Survey of Professional Forecasters), the risk-free rate, as well as quarterly dummy variables as controls for seasonality. Panel B shows the same regressions, but for forecasting log spot price changes. Heteroskedasticity and autocorrelation-adjusted standard errors are used—Newey and West, 1987; three lags). In the pooled regressions, in the two right-most columns, Rogers (1983, 1993) standard errors are used to control for heteroskedasticity, cross-correlations, and autocorrelation (three lags). * Denotes significance at the 10% level, ** denotes significance at the 5% level, and *** denotes significance at the 1% level.

Regressor	Crude Oil		Heating Oil		Gasoline		Natural Gas		All	
	EDF	Zm	EDF	Zm	EDF	Zm	EDF	Zm	EDF	Zm
Panel A: Dependent variable: next-quarter futures return ($r^i_{t,t+1}$)										
$DefRisk_t$	0.058**	0.045***	0.047**	0.035**	0.040*	0.031	0.061*	0.058	0.046**	0.038**
	(0.023)	(0.017)	(0.021)	(0.017)	(0.026)	(0.024)	(0.035)	(0.039)	(0.019)	(0.016)
Controls?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	10.6%	9.7%	9.6%	8.5%	15.2%	13.4%	10.0%	11.0%	7.6%	7.1%
N	107	110	122	125	100	103	79	82	408	420
Panel B: Dependent variable: next-quarter log spot price change ($\Delta s^i_{t,t+1}$)										
$DefRisk_t$	0.056**	0.043**	0.038**	0.033**	0.051**	0.045**	0.063**	0.059*	0.045**	0.038**
	(0.023)	(0.018)	(0.019)	(0.016)	(0.023)	(0.022)	(0.027)	(0.034)	(0.016)	(0.016)
Controls?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	18.0%	16.9%	19.2%	19.1%	17.7%	17.0%	14.4%	15.4%	10.4%	10.1%
N	107	110	122	125	100	103	79	82	408	420

Figure 5: Default risk predicts next-quarter futures returns and spot-price changes.

Read: Table 3 reports regressions of next-quarter futures returns and spot-price changes on lagged producer default risk, with controls. Coefficients on default risk are generally positive and significant. The pooled coefficients are around 0.038 to 0.046 for futures returns and 0.038 to 0.045 for spot-price changes.

Economic message. A one-standard-deviation increase in producer default risk predicts an economically large increase in expected futures returns and expected spot-price changes. This is the main aggregate evidence for the model.

5.5 Table 4: hedgers versus non-hedgers and out-of-sample performance

Table 4
Sample partitions: hedgers vs. non-hedgers and out-of-sample analysis.

Panel A shows the results from joint regressions of Crude Oil, Heating Oil, Gasoline, and Natural Gas futures returns on the aggregate Zmijewski- and EDF-scores of oil producers that hedge and oil producers that do not hedge. The classification of hedgers versus non-hedgers is based on the quarterly reports of firms with SIC code 1311 over the period 2000Q4–2010Q4, as explained in the main text. The default-scores of hedgers are orthogonalized with respect to the non-hedger scores, and all default risk measures are normalized to have unit variance. β_H is the regression coefficient on the measures of hedger default risk, while β_{NH} is the default risk of the producers that do not hedge. Panel B shows out-of-sample R^2's for futures return forecasting regressions using the default risk measures as the predictive variable for each commodity. The in-sample estimation period uses data up until 2005Q4, while the out-of-sample results are for the period 2006Q1–2010Q4 (2010Q1 in the case of the EDF-score given its data availability). The out-of-sample R^2's are calculated with respect to the respective commodity's sample mean up until 2005Q4, as explained in the main text. Heteroskedasticity and autocorrelation-adjusted (Newey-West; three lags) standard errors are used. ** Denotes significance at the 5% level, and *** denotes significance at the 1% level.

Panel A: Hedgers vs. non-hedgers

	Zm-score		EDF-score	
Hedger default risk (β_H)	0.052**	0.096***	0.069***	0.081***
	(0.022)	(0.028)	(0.024)	(0.023)
Non-hedger default risk (β_{NH})	−0.017	−0.004	−0.032	−0.021
	(0.013)	(0.020)	(0.029)	(0.037)
Controls?	No	Yes	No	Yes
R^2	5.6%	11.9%	10.3%	13.9%
N	172	172	160	160

Panel B: Out-of-sample results

	Crude Oil		Heating Oil		Gasoline		Natural Gas	
$(r^i_{t,t+1} = \alpha^i + \beta^i DefRisk_t + \epsilon^i_{t,t+1})$	EDF	Zm	EDF	Zm	EDF	Zm	EDF	Zm
In-sample: until 2005Q4								
$R^2_{in\text{-}sample}$	1.3%	2.6%	1.5%	2.5%	0.9%	0.8%	2.4%	2.1%
$N_{in\text{-}sample}$	90	90	105	105	83	83	62	62
Out-of-sample: 2006Q1–2009Q4/2010Q4								
$R^2_{out\text{-}of\text{-}sample}$	3.5%	6.8%	1.0%	6.2%	7.8%	5.1%	−28.7%	3.9%
$N_{out\text{-}of\text{-}sample}$	16	20	16	20	16	20	16	20

Figure 6: Hedger partition and out-of-sample forecasting evidence.

Read: Panel A shows that the default risk of hedging producers predicts futures returns, while the default risk of non-hedging producers does not. Panel B shows generally positive out-of-sample R^2 values for the 2006Q1-2010Q4 period, with the main exception being EDF for natural gas.

Economic message. The hedger partition strengthens the interpretation that the default-risk effect works through hedging demand, not through default risk in general. The out-of-sample exercise suggests that the relation is not only an in-sample overfit.

5.6 Table 5 and Figure 4: interaction with volatility and speculator capital

Table 5
Interaction between hedging desire and speculator risk tolerance.

The two first columns of Panel A show the results from pooled regressions of Crude Oil, Heating Oil, Gasoline, and Natural Gas futures returns on lagged default risk of oil and gas producers, as measured by the Expected default frequency (EDF) and Zmijewski-score (Zm), respectively, as well as realized variance (RV) for each of the futures returns and their interaction. The two last columns of Panel A show the change in the log spot prices regressed on the same regressors. The controls in the regressions are lagged futures basis, aggregate default risk, forecasted GDP growth, the risk-free rate, as well as quarterly dummy variables (seasonality). Panel B uses growth in broker-dealer assets relative to household assets (BD) as the regressor instead of the realized variance used in Panel A. Panel C has both lagged BD and RV as regressors, as well as the interaction of both of these variables with the default risk measures. Rogers (1993) standard errors are used to control for heteroskedasticity, cross-correlation, and autocorrelation (three lags). * Denotes significance at the 10% level, ** denotes significance at the 5% level, and *** denotes significance at the 1% level.

Regressors	Joint tests across all commodities								
	Futures return		Spot change			Futures return		Spot change	
	EDF	Zm	EDF	Zm		EDF	Zm	EDF	Zm
Panel A					*Panel B*				
RV_t	−0.002	−0.007	0.000	−0.006	BD_t	−0.046***	−0.041**	−0.048***	−0.043***
	(0.013)	(0.012)	(0.015)	(0.012)		(0.011)	(0.015)	(0.010)	(0.015)
$DefRisk_t$	0.036**	0.033*	0.036**	0.034*	$DefRisk_t$	0.032**	0.024	0.031*	0.027
	(0.016)	(0.018)	(0.016)	(0.018)		(0.015)	(0.020)	(0.015)	(0.020)
$RV \times DefRisk$	0.022**	0.015**	0.026**	0.021***	$BD_t \times DefRisk_t$	−0.028***	−0.015**	−0.026***	−0.013
	(0.010)	(0.07)	(0.010)	(0.006)		(0.007)	(0.006)	(0.008)	(0.011)
Controls?	Yes	Yes	Yes	Yes	Controls?	Yes	Yes	Yes	Yes
R^2	8.6%	7.7%	11.8%	11.5%	R^2	13.7%	10.9%	17.5%	15.0%
N	404	416	404	416	N	408	420	408	420
Panel C									
RV_t	−0.004	−0.006	−0.003	−0.006					
	(0.013)	(0.012)	(0.013)	(0.012)					
BD_t	−0.045***	−0.042**	−0.049***	−0.045***					
	(0.011)	(0.016)	(0.010)	(0.016)					
$DefRisk_t$	0.029*	0.018	0.029	0.020					
	(0.017)	(0.023)	(0.018)	(0.024)					
$RV \times DefRisk$	0.017*	0.017**	0.020**	0.021***					
	(0.010)	(0.007)	(0.010)	(0.007)					
$BD \times DefRisk$	−0.026***	−0.014**	−0.024***	−0.012					
	(0.006)	(0.007)	(0.008)	(0.011)					
Controls?	Yes	Yes	Yes	Yes					
R^2	13.8%	11.3%	18.1%	16.1%					
N	404	416	404	416					

Figure 7: Interactions between producer hedging demand, volatility, and speculator risk tolerance.

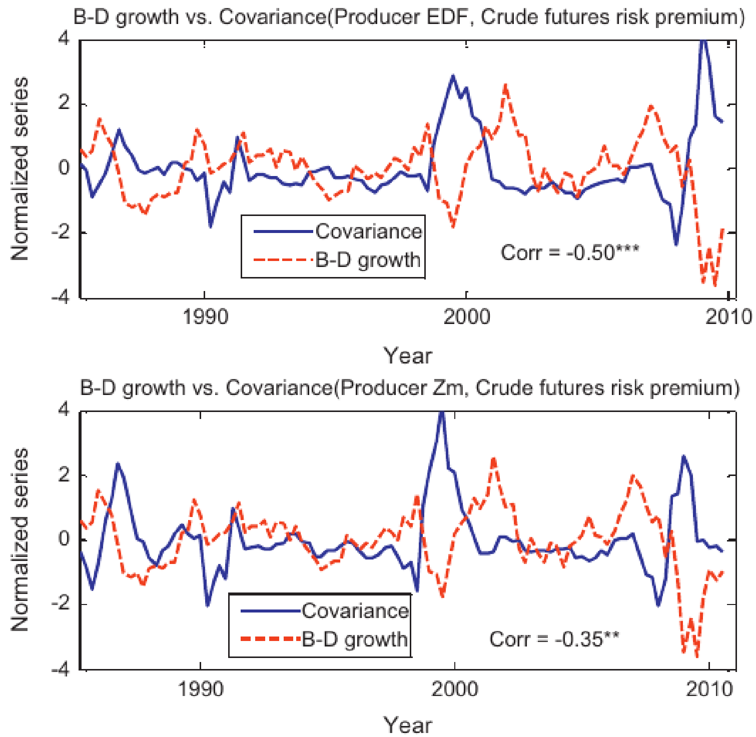

Fig. 4. Hedging versus arbitrage activity. The top plot shows the product of Crude Oil quarterly futures return and lagged demeaned EDF (solid line) versus the lagged growth in broker-dealer assets relative to household assets (dotted line). The former is a measure of the realized covariance of the futures risk premium and hedging demand, whereas the latter is a measure of arbitrage risk tolerance. The bottom plot shows the same for the Zmijewski-score. The two measures are plotted as annual moving averages, and all series are normalized to have zero mean and unit variance. The graphs show that if speculators have high risk tolerance (i.e., B-D growth is high), changes in hedging demand have a small effect on the risk premium as demand is met elastically. If, however, speculators have low risk capacity (B-D growth is low), small changes in hedging demand has a large impact on the futures risk premium as these investors now need more incentive (in the case of increased hedging demand) to take the opposing positions. The sample period is 1983Q3–

Figure 8: Visual evidence on the interaction between hedging demand and arbitrage capital.

Read: Table 5 shows that the interaction between realized variance and default risk is positive, while the interaction between broker-dealer balance-sheet growth and default risk is negative. Figure 4 shows that the covariance of producer default risk with crude-oil futures risk premia is negatively related to broker-dealer growth.

Economic message. The default-risk effect is strongest when futures risk is high and speculator capital is tight. This is exactly the pattern predicted by a downward-sloping speculator demand curve.

5.7 Table 6: CFTC positions, inventory, and basis

Table 6
CFTC positions, inventory, and the basis spread.

This table shows the results from pooled regressions of net short hedger positions relative to total hedger positions as recorded by the CFTC, inventory, and the spot-futures basis for Crude Oil, Heating Oil, Gasoline, and Natural Gas on contemporaneous default risk of oil and gas producers. The default risk measures are the Expected default frequency (EDF) and Zmijewski-score (Zm). The inventory and basis regressions apply an instrumental variable approach described Sections 5.11 and 5.12. The controls in the regressions are the futures basis, aggregate default risk, forecasted GDP growth, the risk-free rate, the lagged dependent variable, as well as quarterly dummy variables (seasonality). Rogers standard errors are used to control for heteroskedasticity, cross-correlation, and autocorrelation (three lags). ** Denotes significance at the 5% level, and *** denotes significance at the 1% level.

Regressors	CFTC positions		Inventory		Basis spread	
	EDF	Zm	EDF	Zm	EDF	Zm
DefRisk	0.097**	0.152***	−0.096**	−0.067**	0.138**	0.151
	(0.048)	(0.044)	(0.047)	(0.038)	(0.069)	(0.093)
Controls?	Yes	Yes	Yes	Yes	Yes	Yes
R^2	16.3%	20.1%	43.0%	44.8%	9.1%	10.1%
N	354	370	408	420	408	420

Figure 9: Default risk, hedger positions, inventory, and basis spread.

Read: Higher producer default risk is associated with more net short hedger positions, lower inventories, and a higher basis spread. The inventory effect is negative and significant for both default-risk measures.

Economic message. Table 6 connects the asset-pricing evidence to the physical mechanism. Producers under distress appear to hedge more and carry less inventory, consistent with financial hedging frictions affecting real commodity supply.

5.8 Table 7: managerial risk aversion

Table 7

Sample partitions: the role of managerial risk aversion.

The table shows results relating firm-level hedging data collected from 10-Q reports for all firms with SIC code 1311 (Crude Oil and Natural Gas Extraction) in the period 2000Q1–2010Q4, to data on executive compensation from ExecuComp from 1992 to 2010. Based on the CEO company stock holdings and option positions relative to the base salary, firms are classified as in a yearly cross-sectional sort as having a "risk-averse" or "risk-tolerant" manager, coded with values of "1" and "0," respectively. High stock holdings and low option holdings leads to classification as a risk-averse manager, while low stock holdings and high option holdings lead to classification as a risk-tolerant manager. The ExecuComp data are annual and Panel A show panel regressions where the dependent variable is the average firm-year data point corresponding to the question "Does the firm currently have significant short crude oil derivatives positions?" and where the binary variable giving the risk-aversion of the manager is on the right-hand side (β_{execu}). Firm size is used as a control in the regressions (β_{size}), as well as time fixed effects. The standard errors are robust and clustered at the CEO level. Panel B shows the results from pooled regressions of Crude Oil, Heating Oil, Gasoline, and Natural Gas futures returns on lagged measures of aggregate default risk of oil and gas producers that have risk-averse managers (β_{RA}) versus the default risk of oil producers that have risk-tolerant managers (β_{RT}). The median Zmijewski- and EDF-scores for each set of firms are used as the forecasting variables. Rogers (1993) standard errors are used to control for heteroskedasticity, cross-correlation, and autocorrelation (three lags). * Denotes significance at the 10% level, and ** denotes significance at the 5% level.

Panel A: Crude oil hedging vs. managerial risk-aversion

Indep.var: Indicator for top half firm equity and bottom half options/salary

Specification	β_{execu}	(S.E.)	β_{size}	(S.E.)	Time fixed effect?	R^2_{adj}	N
1	0.172	(0.106)			No	6.3%	291
2	0.191*	(0.116)			Yes	17.4%	291
3	0.187**	(0.092)	–0.086**	(0.038)	No	23.1%	291
4	0.206**	(0.104)	–0.084**	(0.042)	Yes	29.2%	291

Panel B: Managerial risk aversion and default risk

	Zm-score		EDF-score	
Risk-averse managers' default risk (β_{RA})	0.048*	0.044*	0.016	0.032**
	(0.026)	(0.026)	(0.022)	(0.014)
Risk-tolerant managers' default risk (β_{RT})	–0.007	0.009	0.004	0.027
	(0.038)	(0.040)	(0.039)	(0.034)
Controls?	No	Yes	No	Yes
R^2	7.7%	11.0%	6.0%	11.6%
N	272	272	272	272

Figure 10: Managerial risk aversion and hedging behavior.

Read: Firms with managers classified as more risk-averse are more likely to hedge. The default risk of firms with risk-averse managers predicts futures returns, while the default risk of firms with risk-tolerant managers does not.

Economic message. This table supports the behavioral and corporate-finance foundation of the model: managers with greater downside exposure hedge more, and the pricing effect is concentrated where managerial hedging incentives are strongest.

6 Short summary

- The paper studies how producer hedging demand and constrained speculator capital jointly determine commodity futures risk premia and spot prices.

- The theory combines storage, hedging, and limits to arbitrage. Producers short futures to hedge future production; speculators take the long side but have limited risk-bearing capacity.
- The main pricing equation decomposes the futures risk premium into a standard covariance term and a hedging-pressure term.
- Producer default risk is the empirical proxy for fundamental hedging demand. Higher default risk means stronger desire to hedge.
- EDGAR hedging disclosures validate this proxy: higher default-risk producers are more likely to report significant short crude-oil derivative positions.
- In aggregate data, producer default risk predicts next-quarter futures returns and next-quarter spot-price changes.
- The effect is stronger when futures volatility is high and when broker-dealer balance sheets shrink.
- The results survive controls for business-cycle variables, commodity basis, realized variance, and seasonality.
- The default risk of hedgers predicts returns; the default risk of non-hedgers does not.
- Inventory and CFTC position results line up with the mechanism: higher default risk is associated with more short hedger positions and lower inventories.

7 Conclusion

Acharya, Lochstoer, and Ramadorai show that limits to arbitrage in futures markets can create limits to hedging for commodity producers. When producers want to hedge more and speculators are constrained, futures prices fall relative to expected future spot prices. This raises futures risk premia.

The important additional insight is that the friction spills over into the real side of commodity markets. Expensive hedging changes producer inventory decisions. Producers carry less inventory, which depresses current spot prices relative to expected future spot prices. Therefore, a financial friction in the futures market can affect spot commodity prices.

The paper's empirical design is built around this mechanism. Producer default risk predicts hedging in micro data, predicts futures and spot returns in aggregate data, predicts more strongly when arbitrage capital is tight, and is linked to positions and inventory. The overall message is that commodity risk premia and spot price dynamics cannot be fully understood without considering the balance sheets and incentives of both hedgers and speculators.